

2024



Generating Actionable Insights on SDoH Using AWS with The Center for Health Affairs

Learn how The Center built the SDoH Innovation Hub to address health disparities, structural racism, and poverty using AWS.

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Uses insights

to identify root causes of health disparities, structural racism, and poverty

Ingested data

from over 15 million records and involved over 200 partner organizations

Scales

to meet workloads and needs

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retain control of data

Helps measure

the effectiveness of targeted interventions

Overview

As a healthcare nonprofit, [The Center for Health Affairs](#) (The Center) strives to address health disparities and structural racism. According to the [US Census Bureau](#), Cleveland, Ohio—where The Center has its headquarters—had more than 114,000 people living in poverty in 2019, the highest poverty rate among large US cities. Economic and social challenges, such as housing and food insecurity, impact health outcomes and are known as [social determinants of health](#) (SDoH).

In 2022, The Center and Amazon Web Services (AWS) teamed up to analyze how SDoH contribute to health disparities, with the goal of using data to address root causes. With support from AWS and [IPC Global](#), an [AWS Partner](#), The Center built its SDoH Innovation Hub (SDoH Hub) on AWS to ingest data and strategize solutions, starting with a pilot to address firearm-related deaths in Cleveland before expanding to other communities.



Opportunity | Using Amazon SageMaker to Identify and Address Health Disparities for The Center

Established in 1916, The Center serves as a regional association for hospitals and healthcare organizations, supporting communities in Northeast Ohio.



According to a [study](#) published in the American Journal of Preventive Medicine, a variety of factors impact patient health, with a mere 16 percent of outcomes being determined by clinical care. In other words, socioeconomic factors, health behaviors, and physical environments have a larger combined effect on patient health than medical treatment. AWS and The Center started working together in July 2022 as part of the Impact Computing organization from AWS to bring advanced computing solutions to solve for health equity–related challenges around the world. By December 2023, The Center built the SDoH Hub with support from IPC Global.

The Center previously analyzed the impact of SDoH using [Healthy Northeast Ohio](#), its repository of population health data from publicly available sources. The SDoH Hub built on The Center’s prior work by offering actionable data to inform policy recommendations. “Because we can use the full stack of the AWS Cloud in conjunction with data that’s available, we can do much more and be people-oriented, making someone’s life better rather than just giving them data to read,” says Endrit Meta, data and technology director for The Center. Using [Amazon SageMaker](#)—which provides the fully managed infrastructure, tools, and workflows for building, training, and deploying machine learning models for any use case—The Center can access insights quickly and focus on next steps.

Solution | Ingesting Data from 15 Million Records and Generating Actionable Insights Using AWS

The Center’s SDoH Hub involves over 200 organizations, pulling data from various sources, including publicly available data, to facilitate meaningful, data-driven insights. Over its first 18 months, the SDoH Hub accessed over 15 million records, securely processing both structured and unstructured data. The Center stores data using [Amazon Simple Storage Service](#) (Amazon S3), an object storage service offering industry-leading scalability, data availability, security, and performance. The Center uploads data once and then monitors the system for modifications. To perform data analytics, The Center uses [Amazon Redshift](#), which helps power data-driven decisions using the best price-performance cloud data warehouse. “We can be very agile using AWS,” says Meta. “We built something scalable that supports our different workloads and changing needs. We can scale up and down as needed.”

Along with The Center, there are over 15,000 nonprofits working in Northeast Ohio. Each one has its own mission and methods for addressing a portion of the challenges facing the community. To break down silos, The Center aims to equip its peers with data-informed strategies to address root causes rather than treating symptoms. Using AWS, The Center will also collect and ingest confidential, individual-level data from partner organizations in a secure manner, complying with HIPAA regulations. With the SDoH Hub, organizations retain control of their data, receive targeted intervention recommendations, and can work with the community to take action. “To get guidance, and direction from the community to identify priorities, we built the SDoH Hub on a collaborative and integrated effort to solve protracted local problems,” says Dr. Kim Byas, SDoH leader for The Center.



The Center is focusing on Cleveland for its initial project but has plans to create a template that can be applied globally. From the SDoH Hub's initial data analysis, The Center identified the three zip codes in Cleveland with the highest rates of firearm-related suicide and determined the suicides were correlated with housing insecurity, lack of transportation, and unemployment. The Center plans to analyze data about the individuals living in those zip codes and collaborate with partner organizations to develop policies and programs that can improve access to housing, transportation, and employment. The SDoH Hub will then help assess the impact of these SDoH interventions on future rates of firearm-related suicide in the impacted communities.

Outcome | Expanding and Enhancing the SDoH Hub Using AWS HealthLake

In addition to expanding globally in the future, The Center plans to enhance the SDoH Hub, including building its own models in Amazon SageMaker and migrating data into [AWS HealthLake](#), which helps companies securely store, transform, transact, and analyze health data in minutes. "By connecting data experts from AWS with the healthcare community through The Center, we can drive innovation and offer data-driven solutions," says Byas.

About The Center for Health Affairs

The Center for Health Affairs is a healthcare nonprofit that supports communities and hospitals in Northeast Ohio. Its Social Determinants of Health Innovation Hub helps organizations make data-driven decisions to address health disparities.

AWS Services Used

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon SageMaker

Amazon SageMaker is built on Amazon's two decades of experience developing real-world ML applications, including product recommendations, personalization, intelligent shopping, robotics, and voice-assisted devices.

[Learn more »](#)



AWS HealthLake

AWS HealthLake is a HIPAA-eligible service offering healthcare companies a complete view of individual and patient population health data using FHIR API based transactions to securely store and transform their data into a queryable format at petabyte scale, and further analyze this data using ML models.

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Amazon Redshift

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